

VIETNAM NATIONAL UNIVERSITY – HO CHI MINH CITY
INTERNATIONAL UNIVERSITY
School of Computer Science and Engineering



PROJECT PROPOSAL REPORT (D1)

Course: IT079IU – Principles of Database Management

Semester 1, Academic Year 2026 – 2027

Project Topic 9: Vehicle Registration Authority – Group 7

Instructor: Assoc. Prof. Dr. Nguyen Thi Thuy Loan

TEAM INFORMATION:

No.	Student Name	Student ID	Proposed Role	Email
1	Võ Tuấn Kiệt – Leader	ITCSIU25029	Group Leader	itcsiu25029@student.hcmiu.edu.vn
2	Trần Nguyễn Thành An	ITCSIU25001	Query Developer	itcsiu25001@student.hcmiu.edu.vn
3	Phan Hoàng An	ITCSIU25002	Tester and Writer	itcsiu25002@student.hcmiu.edu.vn
4	Bùi Gia Bảo	ITCSIU25007	Java Developer	itcsiu25007@student.hcmiu.edu.vn
5	Trần Quốc Bảo	ITCSIU25008	Normalization Lead	itcsiu25008@student.hcmiu.edu.vn
6	Trần Khánh Duy	ITCSIU25014	Java Developer	itcsiu25014@student.hcmiu.edu.vn
7	Nguyễn Quang Khải	ITCSIU25019	Database Developer	itcsiu25019@student.hcmiu.edu.vn
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10	Võ Nguyễn Quang Minh	ITCSIU25034	Normalization Lead	itcsiu25034@student.hcmiu.edu.vn
11	Võ Nhật Nam	ITCSIU25038	Database Modeler	itcsiu25038@student.hcmiu.edu.vn
12	Nguyễn Khả Toàn Tấn	ITCSIU25046	Java Developer	itcsiu25046@student.hcmiu.edu.vn
13	Phùng Anh Tuấn	ITCSIU25050	Database Modeler	itcsiu25050@student.hcmiu.edu.vn

Part 1. Abstract:

This project focuses on designing and implementing a comprehensive relational database system for a Vehicle Registration Authority. The primary objective is to develop a centralized data model that tracks vehicle ownership history over time and seamlessly links each vehicle to its registrations, insurance policies, technical inspections, and traffic violations. By integrating these data points, the system enables authorities to efficiently verify full legal and safety compliance from a single unified record. Furthermore, the model addresses complex entity relationships—such as chronological multi-owner tracking, hierarchical owner categorization, and violation-linked fine enforcement—ensuring robust data integrity and formal normalization to 3NF/BCNF. The project also delivers an end-to-end Java desktop application with a JDBC layer to facilitate data entry, strict constraint validation, and advanced administrative querying.

Part 2. Task Allocation Table

Week	Name	Role	Responsibility
W1-3	Phùng Anh Tuấn Võ Nhật Nam	Data modeller	Analyze requirements and define business rules for vehicles, owners, and violations
	Võ Tuấn Kiệt	Group leader	Create project schedule, set internal milestones, and submit Proposal Report (D1)
W3-4	Phùng Anh Tuấn Võ Nhật Nam	Data modeller	Design the Entity-Relationship (ER) diagram with constraints on ERDPlus
	Bùi Gia Bảo Trần Khánh Duy Nguyễn Khả Toàn Tấn	Java developer	Set up project architecture, Maven dependencies, and design GUI wireframes/mockups
W5-6	Phùng Anh Tuấn Võ Nhật Nam	Data modeller	Map ER diagram to relational schema; define primary and foreign keys (Deliverable D2)
	Nguyễn Quang Khải Phan Trần Nguyên Khang	Database developer	Write MySQL CREATE TABLE scripts, constraints (CHECK, UNIQUE, NOT NULL, ON DELETE)
W7-8	Trần Quốc Bảo Võ Nguyễn Quang Minh	Normalization lead	Identify functional dependencies, compute candidate keys, and prove 3NF/BCNF decomposition
	Nguyễn Quang Khải Phan Trần Nguyên Khang	Database developer	Populate 15+ realistic sample data rows per table and configure transaction scripts
	Trần Nguyễn Thành An Võ Đăng Khoa	Query developer	Develop 12 required SQL queries and translate 4 queries into Relational Algebra

W8-9	Bùi Gia Bảo Trần Khánh Duy Nguyễn Khả Toàn Tấn	Java developer	Implement JDBC connection layer, DAO classes with PreparedStatement, and build 5 UI forms
W9-10	Phan Hoàng An	Tester and writer	Conduct software testing, input validation, SQL injection checks, and compile Final Report (D3)
W10	Võ Tuấn Kiệt	Group leader	Perform final review of D3 report, D5 SQL script, D6 Java application, and submit on Blackboard
W11-12	Trần Nguyễn Thành An Võ Đăng Khoa Phan Hoàng An	Query & QA team	Prepare query test cases, optimize database indexes, and draft user manual / README
W13-14	Bùi Gia Bảo Trần Khánh Duy Nguyễn Khả Toàn Tấn	Java developer	Prepare live application demonstration scenario and test database edge cases
W13-15	All members		Design presentation slides (D4), rehearse project demonstration, and conduct final Q&A defense

Part 3. Tools and Programming Languages

Category	Tool	Version	Purpose in Project
Programming Language	Java	OpenJDK 21/25 (LTS)	Core application programming language.
GUI Framework	Java Swing	Standard Java SE	Desktop graphical user interface form.
Database Connector	JDBC	mysql-connector-j-8.3.0	Secure database-to-Java communication layer.
Database Engine	MySQL Server	Version 8.0+	Relational Database Management System (RDBMS).
Database Client	DBeaver Community	Latest Edition	SQL script execution, schema reserve-engineering.
ER Modeling Tool	ERDPlus	Web-based	Conceptual ER Diagram and crow's-foot notation modeling.
Build & Dependency Tool	Apache Maven	Version 3.9+	Managing JDBC dependencies and packaging JAR
Version Control	Git & GitHub	Git 2.4+	Team collaboration, issue tracking, and code reviews.
Documentation & Typesetting	Microsoft Word	Microsoft 365	Compiling D1, D2, and D3 reports into formal PDF.